

Table 10: MMLU [30:40] 5-shot individual task performance.

		MMLU																			
		High School US History		High School World History		Human Aging		Human Sexuality		International Law		Jurisprudence		Logical Fallacies		Machine Learning		Management		Marketing	
Model		Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT
8B	Flan-PaLM	72.7	54.5	57.7	50.0	56.5	47.8	66.7	58.3	76.9	53.8	72.7	36.4	61.1	61.1	45.5	45.5	81.8	36.4	68.0	68.0
	+ Data intervention	59.1	50.0	61.5	53.8	56.5	56.5	58.3	41.7	76.9	38.5	54.5	45.5	61.1	61.1	36.4	27.3	81.8	54.5	76.0	60.0
62B	Flan-PaLM	81.8	72.7	80.8	69.2	60.9	65.2	75.0	50.0	84.6	69.2	63.6	54.5	61.1	66.7	27.3	27.3	81.8	90.9	72.0	68.0
	+ Data intervention	72.7	59.1	65.4	69.2	60.9	56.5	58.3	58.3	84.6	76.9	63.6	36.4	66.7	66.7	36.4	27.3	81.8	90.9	80.0	72.0
62B	Flan-cont-PaLM	81.8	63.6	80.8	84.6	69.6	73.9	66.7	41.7	84.6	84.6	54.5	72.7	72.2	72.2	36.4	36.4	100.0	90.9	84.0	72.0
	+ Data intervention	77.3	68.2	69.2	73.1	78.3	65.2	66.7	50.0	84.6	84.6	63.6	72.7	66.7	72.2	45.5	45.5	100.0	90.9	80.0	80.0
540B	Flan-PaLM	90.9	90.9	84.6	76.9	82.6	82.6	83.3	75.0	92.3	76.9	72.7	72.7	77.8	72.2	45.5	36.4	81.8	90.9	88.0	80.0
	+ Data intervention	90.9	90.9	88.5	80.8	87.0	73.9	75.0	75.0	100.0	76.9	63.6	72.7	72.2	72.2	45.5	54.5	81.8	81.8	88.0	80.0

Table 11: MMLU [40:50] 5-shot individual task performance.

		MMLU																			
		Medical Genetics		Misc.		Moral Disputes		Moral Scenarios		Nutrition		Philosophy		Prehistory		Professional Accounting		Professional Law		Professional Medicine	
Model		Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT
8B	Flan-PaLM	63.6	54.5	68.6	58.1	42.1	36.8	29.0	33.0	54.5	36.4	55.9	52.9	42.9	42.9	35.5	25.8	33.5	31.8	51.6	35.5
	+ Data intervention	81.8	63.6	68.6	61.6	31.6	36.8	29.0	36.0	63.6	36.4	50.0	44.1	51.4	45.7	41.9	45.2	30.0	26.5	41.9	45.2
62B	Flan-PaLM	90.9	90.9	80.2	76.7	65.8	63.2	22.0	46.0	72.7	51.5	64.7	67.6	51.4	60.0	32.3	35.5	47.1	35.3	61.3	71.0
	+ Data intervention	90.9	81.8	74.4	74.4	60.5	73.7	20.0	22.0	72.7	60.6	67.6	64.7	54.3	62.9	38.7	45.2	44.7	30.0	71.0	67.7
62B	Flan-cont-PaLM	90.9	100.0	79.1	79.1	71.1	55.3	24.0	41.0	75.8	60.6	73.5	73.5	74.3	68.6	64.5	45.2	42.4	37.1	64.5	71.0
	+ Data intervention	100.0	100.0	76.7	77.9	60.5	57.9	34.0	34.0	75.8	63.6	73.5	67.6	65.7	74.3	67.7	54.8	41.8	34.1	67.7	67.7
540B	Flan-PaLM	90.9	90.9	82.6	83.7	78.9	60.5	65.0	81.0	84.8	78.8	88.2	73.5	80.0	82.9	51.6	61.3	59.4	51.2	93.5	77.4
	+ Data intervention	90.9	90.9	82.6	86.0	78.9	68.4	73.0	79.0	78.8	75.8	91.2	76.5	82.9	82.9	64.5	61.3	59.4	55.9	93.5	80.6

Table 12: MMLU [50:57] 5-shot individual task performance.

		MMLU															
		Professional Psychology		Public Relations		Security Studies		Sociology		US Foreign Policy		Virology		World Religions		Average	
Model		Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT
8B	Flan-PaLM	46.4	43.5	50.0	41.7	44.4	37.0	68.2	54.5	63.6	45.5	38.9	27.8	78.9	78.9	49.5	39.7
	+ Data intervention	50.7	53.6	50.0	41.7	40.7	29.6	77.3	54.5	72.7	54.5	50.0	16.7	78.9	84.2	48.7	42.8
62B	Flan-PaLM	71.0	66.7	50.0	50.0	70.4	48.1	81.8	68.2	90.9	100.0	55.6	38.9	89.5	84.2	59.8	56.2
	+ Data intervention	71.0	65.2	50.0	50.0	59.3	51.9	77.3	77.3	100.0	100.0	66.7	50.0	89.5	84.2	58.8	56.0
62B	Flan-cont-PaLM	66.7	69.6	58.3	75.0	74.1	59.3	90.9	81.8	100.0	90.9	61.1	44.4	94.7	89.5	65.3	62.9
	+ Data intervention	75.4	72.5	58.3	66.7	59.3	59.3	95.5	81.8	100.0	100.0	72.2	44.4	89.5	89.5	63.7	61.4
540B	Flan-PaLM	76.8	73.9	58.3	50.0	66.7	63.0	100.0	90.9	100.0	100.0	50.0	61.1	84.2	89.5	73.1	69.8
	+ Data intervention	76.8	71.0	58.3	58.3	66.7	66.7	100.0	95.5	100.0	100.0	44.4	55.6	89.5	84.2	72.8	70.2

D.2 BIG-BENCH HARD

BIG-Bench Hard (Suzgun et al., 2022) consists of challenging tasks from BIG-Bench where the model’s performance was better than the average human rater, as reported in Srivastava et al. (2022). In total, there are 23 tasks, two of which have three subtasks (Suzgun et al., 2022). We follow Chung et al. (2022) and Wei et al. (2023) and treat these subtasks as different tasks. Our reported metric in Appendix A.1 and Appendix A.2 is the unweighted average of all subtasks. We use the same prompts as Chung et al. (2022) and Suzgun et al. (2022), which use three few-shot exemplars. Table 13, Table 14, and Table 15 contain model performance on each task in BIG-Bench Hard for Flan-PaLM models before and after synthetic-data intervention.

Table 13: BIG-Bench Hard [:9] individual task performance.

		BIG-Bench Hard																	
		Boolean Expressions		Causal Judgement		Date Understanding		Disambiguation QA		Dyck Languages		Formal Fallacies		Geometric Shapes		Hyperbaton		Logical Deduction Five Objects	
Model		Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT
8B	Flan-PaLM	36.2	44.4	46.8	54.5	60.4	34.0	10.4	39.2	58.0	0.0	15.6	51.6	49.2	4.4	13.6	32.8	62.4	22.0
	+ Data intervention	46.0	48.0	57.8	54.0	16.0	35.2	58.8	40.0	11.2	0.0	48.4	53.2	9.2	4.8	64.4	42.8	32.8	28.0
62B	Flan-PaLM	66.8	74.4	64.7	65.8	43.6	63.6	69.2	26.4	1.6	0.4	55.6	48.8	17.2	16.8	74.8	56.8	53.6	35.6
	+ Data intervention	63.6	67.2	63.1	61.0	44.0	66.0	67.6	60.0	1.2	0.8	52.8	50.8	15.2	14.0	74.4	57.6	50.0	36.8
62B	Flan-cont-PaLM	77.2	82.4	66.3	64.7	52.4	61.2	68.4	68.8	27.2	3.2	55.2	55.2	34.8	22.8	73.2	88.4	52.0	42.0
	+ Data intervention	75.2	81.2	65.2	62.0	51.6	72.8	70.0	59.2	25.6	5.2	59.2	50.0	40.8	33.6	69.6	78.4	54.4	37.2
540B	Flan-PaLM	86.4	81.6	64.2	65.8	59.6	76.8	76.0	65.2	32.0	21.2	60.4	55.2	40.0	42.8	66.0	94.8	55.2	59.2
	+ Data intervention	85.2	84.4	67.9	65.2	60.4	78.4	74.4	70.4	30.0	21.2	61.6	56.0	43.2	43.6	69.6	90.8	54.0	58.0

Table 14: BIG-Bench Hard [9:18] individual task performance.

		BIG-Bench Hard																	
		Logical Deduction Seven Objects		Logical Deduction Three Objects		Movie Recommendation		Multistep Arithmetic		Navigate		Object Counting		Penguins in a Table		Reasoning about Colored Objects		Ruin Names	
Model		Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT
8B	Flan-PaLM	23.6	14.8	25.2	40.0	46.0	46.8	74.4	0.8	0.8	44.4	57.6	29.2	32.0	31.5	30.8	32.8	30.4	28.0
	+ Data intervention	30.8	9.6	47.6	44.8	74.4	44.0	1.2	1.6	58.0	45.6	33.6	42.0	38.4	35.6	32.0	34.0	32.8	16.8
62B	Flan-PaLM	48.4	34.8	73.6	57.6	82.0	73.2	2.0	1.2	61.6	44.4	51.2	48.8	37.0	50.0	50.0	46.4	64.0	48.4
	+ Data intervention	50.0	33.6	72.4	54.0	78.8	80.8	1.6	0.4	60.4	48.0	53.6	54.0	42.5	54.1	46.0	49.2	53.6	40.0
62B	Flan-cont-PaLM	52.0	33.2	70.8	52.0	83.2	84.0	0.8	17.2	62.4	69.6	54.0	68.4	43.2	56.8	50.0	60.4	64.4	74.0
	+ Data intervention	48.4	34.4	70.4	65.6	80.0	84.0	1.2	18.4	61.2	67.2	57.6	56.4	45.9	57.5	53.6	62.8	60.4	60.0
540B	Flan-PaLM	54.0	51.2	86.0	90.0	84.0	86.4	0.8	32.4	67.2	78.4	55.6	87.6	56.8	69.9	67.2	81.2	80.8	63.2
	+ Data intervention	52.8	53.2	87.2	89.6	82.4	86.0	1.2	31.6	67.2	78.4	59.6	88.0	56.2	71.2	64.8	81.2	80.8	64.4

Table 15: BIG-Bench Hard [18:27] individual task performance.

		BIG-Bench Hard																			
		Salient Translation Error Detection		Snarks		Sports Understanding		Temporal Sequences		Tracking Shuffled Objects (5)		Tracking Shuffled Objects (7)		Tracking Shuffled Objects (3)		Web of Lies		Word Sorting		Average	
Model		Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT	Direct	CoT
8B	Flan-PaLM	42.4	0.0	27.2	60.7	69.1	69.6	63.6	25.6	14.4	18.0	18.0	14.8	16.4	32.0	33.2	49.6	51.6	2.0	36.2	30.5
	+ Data intervention	23.6	0.0	62.4	63.5	64.4	67.6	16.8	23.2	18.4	17.2	15.6	14.8	34.8	32.8	51.6	52.4	5.6	1.6	36.5	31.6
62B	Flan-PaLM	44.4	38.4	82.6	83.1	79.2	82.4	31.6	39.6	22.0	23.2	14.8	20.8	22.4	32.8	48.4	89.6	10.4	9.2	47.1	44.9
	+ Data intervention	46.4	44.4	78.7	77.5	78.8	83.2	27.6	44.0	21.6	18.8	16.4	14.0	23.6	31.6	51.6	93.2	10.4	8.4	46.1	46.1
62B	Flan-cont-PaLM	48.8	42.0	83.1	80.3	82.4	84.0	33.6	67.6	20.0	25.2	19.6	16.4	23.2	37.6	48.8	95.2	16.0	16.0	50.5	54.4
	+ Data intervention	49.6	44.8	80.3	83.7	83.6	86.8	28.0	65.2	20.4	30.8	18.8	21.6	27.6	37.2	47.2	98.0	14.8	17.2	50.4	54.5
540B	Flan-PaLM	54.0	47.6	83.1	75.3	81.6	88.0	76.8	89.2	24.8	49.6	23.2	36.0	32.8	63.2	59.6	100.0	32.8	34.4	57.8	66.2
	+ Data intervention	54.0	55.2	84.3	76.4	83.6	90.4	80.4	91.6	26.4	48.8	23.2	37.2	34.8	64.8	58.0	100.0	33.6	35.6	58.4	67.1

D.3 MMLU (ZERO-SHOT)

In Appendix A.3, we evaluated models on MMLU (Hendrycks et al., 2021) in a zero-shot setting (as opposed to the five-shot setting in Appendix A.1). We show per-task performance results for zero-shot MMLU for Flan-PaLM models before and after synthetic-data intervention in Table 16, Table 17, Table 18, Table 19, Table 20, and Table 21.

Table 16: MMLU [:10] 0-shot individual task performance.

		MMLU									
Model		Abstract Algebra	Anatomy	Astronomy	Business Ethics	Clinical Knowledge	College Biology	College Chemistry	College Comp. Sci.	College Math	College Medicine
8B	Flan-PaLM	27.3	57.1	68.8	36.4	41.4	56.2	37.5	36.4	9.1	45.5
	+ Data intervention	36.4	50.0	43.8	45.5	37.9	62.5	12.5	45.5	36.4	45.5
62B	Flan-PaLM	27.3	64.3	75.0	63.6	55.2	75.0	37.5	63.6	36.4	72.7
	+ Data intervention	27.3	64.3	56.2	54.5	55.2	75.0	37.5	63.6	63.6	68.2
62B	Flan-cont-PaLM	27.3	64.3	75.0	63.6	75.9	68.8	37.5	54.5	54.5	72.7
	+ Data intervention	36.4	57.1	68.8	63.6	65.5	62.5	37.5	63.6	54.5	81.8
540B	Flan-PaLM	0.0	50.0	75.0	63.6	79.3	81.2	50.0	72.7	36.4	81.8
	+ Data intervention	9.1	50.0	75.0	54.5	79.3	87.5	50.0	63.6	36.4	81.8

Table 17: MMLU [10:20] 0-shot individual task performance.

		MMLU									
Model		College Physics	Computer Security	Conceptual physics	Econometrics	Electrical Engineering	Elementary Mathematics	Formal Logic	Global Facts	High School Biology	High School Chemistry
8B	Flan-PaLM	54.5	54.5	38.5	25.0	56.2	29.3	28.6	50.0	43.8	22.7
	+ Data intervention	45.5	36.4	53.8	16.7	50.0	29.3	14.3	10.0	40.6	40.9
62B	Flan-PaLM	72.7	54.5	53.8	50.0	43.8	39.0	35.7	30.0	68.8	31.8
	+ Data intervention	45.5	54.5	53.8	41.7	56.2	39.0	7.1	20.0	59.4	22.7
62B	Flan-cont-PaLM	63.6	63.6	61.5	50.0	50.0	53.7	28.6	40.0	68.8	31.8
	+ Data intervention	45.5	63.6	50.0	58.3	56.2	56.1	35.7	30.0	62.5	31.8
540B	Flan-PaLM	72.7	63.6	69.2	58.3	81.2	51.2	50.0	50.0	75.0	59.1
	+ Data intervention	81.8	81.8	69.2	58.3	75.0	58.5	28.6	40.0	78.1	63.6

Table 18: MMLU [20:30] 0-shot individual task performance.

		MMLU									
Model		High School Comp. Sci.	High School European History	High School Geography	High School Gvmt & Politics	High School Macroeconomics	High School Math	High School Microeconomics	High School Physics	High School Psychology	High School Statistics
8B	Flan-PaLM	33.3	66.7	68.2	61.9	44.2	27.6	61.5	47.1	65.0	39.1
	+ Data intervention	33.3	83.3	63.6	61.9	41.9	44.8	53.8	41.2	66.7	30.4
62B	Flan-PaLM	55.6	88.9	81.8	76.2	62.8	20.7	69.2	29.4	88.3	47.8
	+ Data intervention	55.6	94.4	86.4	71.4	62.8	31.0	65.4	29.4	86.7	52.2
62B	Flan-cont-PaLM	55.6	88.9	90.9	81.0	62.8	24.1	88.5	29.4	93.3	60.9
	+ Data intervention	55.6	83.3	86.4	76.2	62.8	34.5	76.9	17.6	90.0	56.5
540B	Flan-PaLM	100.0	77.8	95.5	95.2	79.1	27.6	96.2	17.6	95.0	73.9
	+ Data intervention	88.9	77.8	95.5	95.2	79.1	24.1	92.3	11.8	95.0	69.6